Concentration of Random Tensor Sumsets

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Abstract

Cutler, Pebody and Sarkar introduced a fractional size $\|\gamma\|$ for finitely supported weights on $\mathbb{Z}$ and showed that if $A_n \subset \mathbb{Z}^n$ includes each point independently with probability $\prod_i \alpha(x_i)$, for a fractional set $\alpha : \mathbb{Z} \to [0,1]$ of total mass $a > 1$, then $\mathbb{E}|A_n + kA_n|$ grows at the exponential rate $\|\alpha + k\alpha\|$ for fixed nonzero k. They conjectured that the random cardinality itself has this rate, and proved it only in a strictly spartan regime. For every such α and every fixed nonzero integer k we prove $\operatorname{Var}|A_n+kA_n| \le 4(a^n+2)\,\mathbb{E}|A_n+kA_n|$ and $\|\alpha+k\alpha\| > a$; hence $|A_n+kA_n|/\mathbb{E}|A_n+kA_n| \to 1$ almost surely and in L^2, with exponentially small probability of any fixed relative deviation, and $\frac{1}{n}\log|A_n+kA_n| \to \log\|\alpha+k\alpha\|$ almost surely, with no spartan assumption. The variance bound is a direct deletion-counting argument combined with the Efron–Stein inequality and a covariance identity for Bernoulli coordinates. An explicit example has its minimizing exponent strictly inside $(0,1)$, and an example at mass one shows that $a > 1$ is needed. Consequences: a single sequence of finite integer sets realizes the fractional dilate exponents for every fixed nonzero k, and the closures of finite-set and fractional-set nonzero-dilate profiles coincide. We also note that the simultaneous-realization conjecture as printed fails at $k = 0$, and that the fractional version of the question $|A||A + 2A| \le |A + A|^2$ fails below mass one and is equivalent to the finite question above mass one. Limitations: only fractional sets (weights at most one) and nonzero dilates are covered; the expectation asymptotics are imported from Cutler–Pebody–Sarkar; the profile regions themselves are not determined. Small exact enumerations are supplied as sanity checks.

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Problem and contribution

For a finitely supported $\gamma : \mathbb{Z} \to [0,\infty)$, Cutler, Pebody and Sarkar [1] define the size

$$\|\gamma\| = \inf_{0\le p\le 1} \sum_{t\in\operatorname{supp}(\gamma)} \gamma(t)^p.$$

A *fractional set* is a finitely supported $\alpha : \mathbb{Z} \to [0,1]$; for it the infimum is attained at $p = 1$, so $\|\alpha\| = \sum_t \alpha(t)$. For an integer k put $(\alpha + k\alpha)(t) = \sum_{u+kv=t} \alpha(u)\alpha(v)$. A random set $A_n \subset \mathbb{Z}^n$ is *drawn from* α^n if distinct $x = (x_1, \dots, x_n)$ are included independently with probability

$$q_x = \prod_{i=1}^{n} \alpha(x_i). \qquad (1)$$

We use one result of [1], with its hypotheses as stated there and its proof.

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We use the convention $\log 0 = -\infty$. Conjecture 44 of [1] asks whether $\frac{1}{n}\log|A_n + kA_n|$ itself converges to $\log\|\alpha + k\alpha\|$ (stated there for positive k and for "fractional dilates", although the sampling rule (1) is a probability model only for fractional sets). The authors of [1] prove the typical behaviour only in a "strictly spartan" regime (their Theorem 5), where collisions are sparse enough to compare with $|A_n|^2$. Outside that regime the question is a lower-tail one: the expectation might be carried by rare sets with unusually large images.

Results of this paper.

1. (Theorem 1) For every fractional set with $\|\alpha\| > 1$ and every integer $k \neq 0$, $\operatorname{Var}|A_n + kA_n| \le 4(\|\alpha\|^n + 2)\,\mathbb{E}|A_n + kA_n|$ and $\|\alpha + k\alpha\| > \|\alpha\|$; consequently $|A_n + kA_n|/\mathbb{E}|A_n + kA_n| \to 1$ almost surely and in L^2, with exponentially

small probability of any fixed relative deviation. In particular $\frac{1}{n}\log|A_n+kA_n| \to \log\|\alpha+k\alpha\|$ almost surely. This settles Conjecture 44 of [1] in its well-posed fractional-set form, without any spartan hypothesis, and also for negative k. Example 2 exhibits an interior minimizer, outside the strictly spartan regime, and Remark 3 shows that the hypothesis $\|\alpha\| > 1$ cannot be relaxed to ≥ 1.

2. (Corollaries 5 and 6) One sequence of finite subsets of $\mathbb{Z}$ realizing the fractional dilate exponents for every fixed nonzero k, and equality of the closures of finite-set and fractional-set dilate profiles. This is the nonzero-dilate, fractional-set case of Conjecture 45 of [1].
3. (Proposition 7, Remark 8) Two boundary observations on [1]: Conjecture 45 as printed includes $k = 0$, where it fails; Question 46(1) as printed fails for every fractional set of mass in $(0, 1)$, and its restriction to mass greater than one is equivalent to Question 14 of [1].

The proofs depend only on 1, Theorem 4 as quoted and on the Efron–Stein inequality [2]. Not claimed: any statement for fractional dilates with some value larger than 1 (for which (1) is not a probability model); an answer to Question 14 of [1] or to the mass-greater-than-one case of its Question 46(1), which are equivalent; explicit convergence rates of $(\mathbb{E}|A_n + kA_n|)^{1/n}$.

Related work and distinction

The expectation asymptotics, the fractional size $\|\cdot\|$, and the strictly spartan concentration result are due to Cutler, Pebody and Sarkar [1]; we use their Theorem 4 as a black box and replace the spartan comparison with $|A_n|^2$ by a direct variance bound valid in all regimes. Concentration of sumsets and images of linear forms of random sets has been studied in homogeneous models: Hegarty and Miller [3] found threshold behaviour for sumsets, difference sets and binary linear forms of sparse binomial subsets of an interval; Jeong and Miller [4] give sharp global and local phase transitions for fixed linear forms in that model; Nenadov and Verlinde [5] determine asymptotically the logarithm of the upper-tail probability $\mathbb{P}(|\mathbb{N} \setminus (A + A)| \geq m)$ for a p-random subset A of $\mathbb{N}$; and the distribution and moments of the number of missing sums at fixed density were studied by Lazarev, Miller and O'Bryant [6], Chu et al. [7] and Jambhale et al. [8]. The random-sumset results compared here all use a single homogeneous inclusion probability on an interval or on $\mathbb{N}$. The model here has tensor-product, point-dependent probabilities on $\mathbb{Z}^n$, and its exponential rate is the fractional size, which is in general not $|A_n|^2$ or $|A_n|$. The novelty claim is relative to the literature we searched: Semantic Scholar, OpenAlex, arXiv and Crossref (October 2026), with queries on Conjecture 44 of [1], fractional dilates, random tensor-product sumsets and works citing [1]. Within that search we found no work proving Conjecture 44 of [1]; this is not an exhaustive survey.

A direct variance bound

Key Lemma (deletion variance bound). Let A be a random subset of a finite $X \subset \mathbb{Z}^d$ including the points $x \in X$ independently with probabilities $p_x \in [0, 1]$, let $k \neq 0$, and put $Z = |A + kA|$, $\lambda = \mathbb{E}|A|$, $\mu = \mathbb{E}Z$. Then

$$\operatorname{Var} Z \leq 4\,\mathbb{E}\big[|A|\,Z\big] \leq 4(\lambda + 2)\mu.$$

Proof. For $x \in A$ let $D_x = Z(A) - Z(A \setminus \{x\})$ be the deletion loss. Fix for each $z \in A + kA$ one representation $z = u + kv$ with $u, v \in A$. If deleting x removes z, then $x \in \{u, v\}$; hence each image point is counted by at most two deletion losses and

$$\sum_{x\in A} D_x \leq 2Z.$$

The points removed by deleting x lie in $(x + kA) \cup (A + kx)$, so $D_x \leq 2|A|$, and therefore $\sum_{x\in A} D_x^2 \leq 4|A|Z$.

We use the Efron–Stein inequality [2] in the general form of Steele [9]: if $F = f(X_1, \dots, X_N)$ with independent $X_1, \dots, X_N$, and $\mathbb{E}_i F$ denotes the expectation over X_i alone with the other coordinates fixed, then $\operatorname{Var} F \leq \sum_{i=1}^{N} \mathbb{E}\big[(F - \mathbb{E}_i F)^2\big]$, with no further constant. Here the coordinates are the indicators $X_x = \mathbf{1}_{\{x\in A\}} \sim \text{Bernoulli}(p_x)$. Let Δ_x be the value of Z with $X_x = 1$ minus its value with $X_x = 0$, the other coordinates fixed; Δ_x is independent of X_x, $Z - \mathbb{E}_x Z = (X_x - p_x)\Delta_x$, and on $\{X_x = 1\}$ we have $\Delta_x = D_x$. Hence

$$\mathbb{E}\big[(Z - \mathbb{E}_x Z)^2\big] = p_x(1 - p_x)\,\mathbb{E}\Delta_x^2 \leq p_x\,\mathbb{E}\Delta_x^2 = \mathbb{E}\big[X_x D_x^2\big],$$

and summing over x gives $\operatorname{Var} Z \leq \mathbb{E}\sum_{x\in A} D_x^2 \leq 4\,\mathbb{E}[|A|Z]$.

For the second inequality, independence of X_x and Δ_x gives

$$\operatorname{Cov}(X_x, Z) = p_x(1-p_x)\,\mathbb{E}\Delta_x = (1-p_x)\,\mathbb{E}[X_x D_x] \le \mathbb{E}[X_x D_x],$$

since $D_x \ge 0$. Summing over $x \in X$,

$$\mathbb{E}\big[|A|Z\big] - \lambda\mu = \sum_{x\in X} \operatorname{Cov}(X_x, Z) \le \mathbb{E}\sum_{x\in A} D_x \le 2\mu. \qquad \square$$

The proof covers diagonal representations $u = v$, $k = -1$, the empty set, and probabilities $p_x \in \{0,1\}$.

Main theorem

Theorem 1. Let $\alpha : \mathbb{Z} \to [0,1]$ be finitely supported with $a := \sum_t \alpha(t) > 1$, let A_n be drawn from α^n, let $k \in \mathbb{Z}\setminus\{0\}$, and put $Z_n = |A_n + kA_n|$, $\mu_n = \mathbb{E}Z_n$ and $b = \|\alpha + k\alpha\|$. Then $b > a$ and

$$\operatorname{Var} Z_n \le 4(a^n + 2)\,\mu_n \qquad (n \ge 1).$$

Consequently, for every $\delta > 0$,

$$\mathbb{P}\big(|Z_n/\mu_n - 1| > \delta\big) \le \frac{4(a^n+2)}{\delta^2\mu_n} = \exp\big(-n\log(b/a) + o(n)\big), \qquad (2)$$

and $Z_n/\mu_n \to 1$ almost surely and in L^2, for any sequence (A_n) on a common probability space in which each A_n has the law (1); no independence between different n is needed. Moreover, for every $\varepsilon > 0$ and every $0 < c < \log(b/a)$ there is n_0 such that for all $n \ge n_0$

$$\mathbb{P}\big(b^n e^{-\varepsilon n} \le Z_n \le b^n e^{\varepsilon n}\big) \ge 1 - e^{-cn},$$

and $\frac{1}{n}\log Z_n \to \log b$ almost surely.

Proof. Step 1: $b > a$. Let $B = \operatorname{supp}(\alpha)$, $m = |B|$ and $\gamma = \alpha + k\alpha$. Since $0 \le \alpha \le 1$, every value satisfies $\gamma(t) = \sum_v \alpha(t-kv)\alpha(v) \le \sum_v \alpha(v) = a$, and $\sum_t \gamma(t) = a^2$. For $0 < p \le 1$,

$$\sum_{t\in\operatorname{supp}(\gamma)} \gamma(t)^p = \sum_t \gamma(t)\,\gamma(t)^{p-1} \ge a^{p-1}\sum_t \gamma(t) = a^{p+1} > a.$$

For $p = 0$ the sum is $|\operatorname{supp}(\gamma)| = |B + kB| \ge |B| + |kB| - 1 = 2m - 1$, because $k \ne 0$ gives $|kB| = m$; and $a > 1$ with $\alpha \le 1$ gives $m \ge 2$ and $m \ge a$, so $2m - 1 > m \ge a$. The finite sum $p \mapsto \sum_t \gamma(t)^p$ is continuous on the compact interval $[0,1]$, so its minimum b is attained and $b > a$.

Step 2: variance and deviations. By the Key Lemma with $X = B^n$ and $\mathbb{E}|A_n| = a^n$ 1, Theorem 4, $\operatorname{Var} Z_n \le 4(a^n+2)\mu_n$. By 1, Theorem 4, $\mu_n = b^{n+o(n)}$, and Chebyshev's inequality gives (2). Since $b > a$, the right-hand side of (2) is summable; the first Borel–Cantelli lemma applied with $\delta = 1/j$ for every $j \ge 1$ gives $Z_n/\mu_n \to 1$ almost surely, and $\mathbb{E}(Z_n/\mu_n - 1)^2 = \operatorname{Var} Z_n/\mu_n^2 \to 0$ gives convergence in L^2. For the last statement take $\delta = 1/2$ in (2): for n large, $[\mu_n/2, 3\mu_n/2] \subseteq [b^n e^{-\varepsilon n}, b^n e^{\varepsilon n}]$ and the bound in (2) is at most e^{-cn}. Finally $\frac{1}{n}\log Z_n = \frac{1}{n}\log\mu_n + \frac{1}{n}\log(Z_n/\mu_n) \to \log b$ almost surely. $\square$

Only the dependence of n_0 on ε is unspecified; the exponential rate c can be any number below $\log(b/a)$, independently of ε.

Example 2 (an interior minimizer). Let $\alpha = \frac{7}{8}\mathbf{1}_{\{0,1\}}$ and $k = 1$. Then $a = \frac{7}{4}$ and $\gamma = \alpha + \alpha$ takes the values $r, 2r, r$ with $r = \frac{49}{64}$, so $f(p) = \sum_t \gamma(t)^p = r^p(2 + 2^p)$. One computes $f'(0) = \log(2r^3) < 0$ and $f'(1) = 2r\log(2r^2) > 0$, since $2r^3 = \frac{235298}{262144} < 1$ and $2r^2 = \frac{4802}{4096} > 1$; and f is a positive combination of exponentials, so $f'' > 0$. Hence the unique minimizer $p_* = \log_2\big(-2\log r/\log(2r)\big)$ lies in $(0,1)$, with $p_* \approx 0.32604$ and $b = f(p_*) \approx 2.98225$. In particular $b < 3 < \frac{49}{16} = a^2$, and Theorem 1 gives $\frac{1}{n}\log\big(|A_n + A_n|/|A_n|^2\big) \to \log\frac{16b}{49} < 0$ almost surely. The strictly spartan condition of [1, Theorem 5] corresponds to $f'(1) < 0$, so this example lies outside it.

Remark 3 (the mass hypothesis). If $a < 1$ then $\sum_n \mathbb{P}(A_n \ne \varnothing) \le \sum_n a^n < \infty$, so A_n is eventually empty almost surely under any coupling. At $a = 1$ take $\alpha = \frac{1}{2}\mathbf{1}_{\{0,1\}}$ and $k = 1$: then $b = 1$, but each of the 2^n points of $\{0,1\}^n$ is included with probability 2^{-n}, so $\mathbb{P}(A_n = \varnothing) = (1-2^{-n})^{2^n} \to e^{-1}$, and the lower bound of Theorem 1 fails with probability bounded away from zero. Thus $a > 1$ cannot be relaxed to $a \ge 1$.

Simultaneous realization

Corollary 5 (simultaneous realization). Let α be a fractional set with $a = \sum_t \alpha(t) > 1$. There are finite sets $S_n \subset \mathbb{Z}$ such that, for every fixed $k \in \mathbb{Z} \setminus \{0\}$,

$$\frac{\log|S_n|}{n} \to \log a, \qquad \frac{\log|S_n + kS_n|}{n} \to \log\|\alpha + k\alpha\|.$$

The convergence is for each fixed k, not uniform in k. Consequently, for every finite nonempty $K \subset \mathbb{Z} \setminus \{0\}$ and every $\varepsilon > 0$ some finite $S \subset \mathbb{Z}$ satisfies

$$\max_{k\in K} \left| \frac{\log|S + kS|}{\log|S|} - \frac{\log\|\alpha + k\alpha\|}{\log\|\alpha\|} \right| < \varepsilon.$$

Proof. Intersect the probability-one events of Theorem 1 over the countably many $k \neq 0$. Also $\mathbb{E}|A_n| = a^n$ and $\operatorname{Var}|A_n| \le a^n$, so Chebyshev bounds $\mathbb{P}(|A_n| < a^n/2)$ by $4a^{-n}$ and Markov bounds $\mathbb{P}(|A_n| > a^n e^{\varepsilon n})$ by $e^{-\varepsilon n}$; by Borel–Cantelli over rational ε, $|A_n| = a^{n+o(n)}$ almost surely. Fix one outcome in all these events. Choose an integer L with $\operatorname{supp}(\alpha) \subseteq [-L, L]$, put $M_n = 2(n+1)L + 2$ and $\phi_n(x) = \sum_{i=1}^{n} x_i M_n^{i-1}$. Since $M_n > 2(n+1)L + 1$, ϕ_n is injective on $[-(n+1)L, (n+1)L]^n$: in a nonzero difference the highest nonzero base-M_n coordinate dominates the lower ones. For fixed k and $n \ge |k|$, all points of A_n and of $A_n + kA_n$ lie in this box, so $S_n = \phi_n(A_n)$ satisfies $|S_n| = |A_n|$ and $|S_n + kS_n| = |A_n + kA_n|$. Dividing the limits gives the second statement. □

For finite nonempty $K \subset \mathbb{Z} \setminus \{0\}$ let $\mathcal{F}_K$ be the closure of $\big(\log|S + kS|/\log|S|\big)_{k\in K}$ over finite $S \subset \mathbb{Z}$ with $|S| \ge 2$, and $\mathcal{F}_K^{\mathrm{frac}}$ the closure of $\big(\log\|\alpha + k\alpha\|/\log\|\alpha\|\big)_{k\in K}$ over fractional sets with $\|\alpha\| > 1$.

Corollary 6 (profile equivalence). $\mathcal{F}_K = \mathcal{F}_K^{\mathrm{frac}}$ for every finite nonempty $K \subset \mathbb{Z} \setminus \{0\}$.

Proof. Corollary 5 gives $\mathcal{F}_K^{\mathrm{frac}} \subseteq \mathcal{F}_K$. Conversely, for $\alpha = \mathbf{1}_S$ with $|S| \ge 2$ we have $\|\alpha\| = |S| > 1$, and every nonzero value of $\alpha + k\alpha$ is a positive integer, so $t \mapsto \gamma(t)^p$ is nondecreasing in p termwise and the infimum is attained at $p = 0$: $\|\alpha + k\alpha\| = |S + kS|$. □

Boundaries of the conjectures in [1]

Conjecture 45 of [1] asserts, for every fractional dilate α, every N and $\varepsilon > 0$, a finite $S \subset \mathbb{Z}$ approximating $\log\|\alpha + k\alpha\|/\log\|\alpha\|$ by $\log|S + kS|/\log|S|$ for all $|k| \le N$. That range contains $k = 0$.

Proposition 7. Let $\alpha = \frac{3}{5}(\mathbf{1}_{\{0\}} + \mathbf{1}_{\{1\}})$. Then $\log\|\alpha + 0\alpha\|/\log\|\alpha\| = 2$, whereas $\log|S + 0S|/\log|S| = 1$ for every finite S with $|S| \ge 2$.

Proof. $\|\alpha\| = 6/5$ and $(\alpha + 0\alpha)(t) = \alpha(t)\sum_j \alpha(j)$ takes the value $18/25$ at two points, so it is a fractional set with size $36/25 = (6/5)^2$. On the other hand $S + 0S = S$. □

So Conjecture 45 is false as printed, independently of the issue that (1) is not a probability model for fractional dilates with values above 1. Corollaries 5 and 6 prove its natural nonzero-dilate statement for fractional sets; the corresponding deterministic statement for arbitrary fractional dilates remains open.

Remark 8 (Question 46(1) of [1]). Question 46(1) asks whether $\|\alpha\|\,\|\alpha + 2\alpha\| \le \|\alpha + \alpha\|^2$ for every finitely supported $\alpha : \mathbb{Z} \to [0, 1]$. If $0 < m := \|\alpha\| < 1$, then every value of $\alpha + \alpha$ and of $\alpha + 2\alpha$ is at most $\sum_u \alpha(u) = m < 1$, so both are fractional sets of total mass m^2; the inequality would read $m^3 \le m^4$, which is false. If $m = 1$ both sides equal 1. If $m > 1$, dividing logarithms by $\log m > 0$ turns the inequality into $1 + r_2 \le 2r_1$ with $r_k = \log\|\alpha + k\alpha\|/\log\|\alpha\|$; this closed condition holds on $\mathcal{F}_{\{1,2\}}^{\mathrm{frac}}$ if and only if it holds on $\mathcal{F}_{\{1,2\}}$ (Corollary 6), that is, if and only if Question 14 of [1] ($|A|\,|A + 2A| \le |A + A|^2$ for all finite A) has a positive answer. Thus fractional and finite formulations of this question transfer exactly, but Corollary 6 does not decide it.

Limitations and scope

The theorem is restricted to fractional sets (values in $[0, 1]$), mass greater than one (Remark 3), and nonzero integer dilates. The relative concentration is around $\mathbb{E}|A_n + kA_n|$, whose growth rate is imported from 1, Theorem 4 only qualitatively, so n_0 is not explicit. Corollary 6 shows that optimizing over fractional sets describes the closure of finite nonzero-dilate profiles exactly, but it does not determine those regions.

Reproducibility

The proofs are self-contained apart from 1, Theorem 4 and [2, 9]. The supplement contains `check_lemmas.py` (Python 3 standard library, a few seconds), which checks by exact rational enumeration the deletion bounds $\sum_{x\in A} D_x \le 2Z$ and $D_x \le 2|A|$, the covariance bound $\mathbb{E}[|A|Z] \le (\lambda+2)\mu$ and the variance bound of the Key Lemma on small instances with nonuniform probabilities including 0 and 1 (subsets of $\{0,1,2\}^d$, $d \le 2$, dilates $k \in \{\pm1, \pm2, 3\}$); the Bernoulli Efron–Stein identity $\mathbb{E}[(F-\mathbb{E}_i F)^2] = p_i(1-p_i)\mathbb{E}\Delta_i^2$; the inequality $b > a$ and the numbers of Example 2; the example of Proposition 7; and the failure of Question 46(1) on random fractional sets of mass below one. These are sanity checks, not proofs.

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